

Question

A student conducted a titration analysis on a sample of $K_3Fe(C_2O_4)_3 \cdot 3H_2O$. The method involved: weighing an appropriate amount of the sample, dissolving it in dilute sulfuric acid, titrating it with $KMnO_4$ under heating until the endpoint was reached, then reducing the above titrated solution with Zn until no color reaction with $KSCN$ was observed. After filtering out the excess Zn , the solution was titrated with $K_2Cr_2O_7$ until the endpoint was reached. Based on the results of these two titrations, the molar ratio of Fe to $C_2O_4^{2-}$ in the sample could be calculated.

The student might have committed certain improper operations during the experiment:

1. The sample was left in the air for too long before titration, leading to efflorescence.
2. The sample was exposed to sunlight before titration, causing a color change.
3. A small amount of the solution splashed out while dissolving the sample.
4. A transparent burette was used instead of a brown one for the $KMnO_4$ titration.
5. Inadequate washing after filtering the Zn .

Assuming no other errors beyond the issues mentioned above, how many of these improper operations could potentially result in an elevated molar ratio value of $n(Fe)/n(C_2O_4^{2-})$?

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

Answer

Correct Answer: **A**

Detailed Explanation

Titration of $C_2O_4^{2-}$ content with $KMnO_4$.

CHECKPOINT

1 PTS

$KMnO_4$ titration of $C_2O_4^{2-}$ content

Titration of Fe^{2+} (i.e., Fe content) with $K_2Cr_2O_7$.

CHECKPOINT

1 PTS

$K_2Cr_2O_7$ titration of Fe content

If the ratio $n(Fe)/n(C_2O_4^{2-})$ is high, it means the actual amount of $KMnO_4$ used is high or the actual amount of $K_2Cr_2O_7$ used is low.

CHECKPOINT

1 PTS

Actual amount of $KMnO_4$ used is high or actual amount of $K_2Cr_2O_7$ used is low

Operation 1 does not affect the ratio. Weathering only affects the water of crystallization and does not influence the redox titration part.

CHECKPOINT

1 PTS

Operation 1 does not affect the ratio

Operation 2 does not affect the ratio.

The photolysis reaction is



Although it superficially reduces the amount of $C_2O_4^{2-}$, since Fe^{2+} is generated, the total electron transfer number remains unchanged during $KMnO_4$ titration, and thus the calculated $C_2O_4^{2-}$ content is unaffected.

CHECKPOINT

5 PTS

Operation 2 does not affect the ratio

Operation 3 does not affect the ratio. If the solution splashes out before titration, the values for both titrations decrease proportionally, leaving the ratio unaffected.

CHECKPOINT

1 PTS

Operation 3 does not affect the ratio

Operation 4 causes the ratio to decrease. Photodecomposition of $KMnO_4$ reduces its concentration, leading to higher titration usage.

CHECKPOINT

1 PTS

Operation 4 causes the ratio to decrease

Operation 5 causes the ratio to decrease. If the residue is not fully washed, Fe^{2+} does not completely enter the next titration step, resulting in a lower measured Fe content.

CHECKPOINT

1 PTS

Operation 5 causes the ratio to decrease

In conclusion, no operation causes the ratio to increase.

CHECKPOINT

1 PTS

0 operations cause the ratio to increase